

■ 1.5 Mathematical Operations in *Mathematica*

■ 1.5.1 Symbolic Mathematics

Mathematica's ability to deal with symbolic expressions, as well as numbers, allows you to use it for many kinds of mathematics.

Calculus is one example. With *Mathematica*, you can differentiate an expression *symbolically*, and get a formula for the result.

This finds the derivative of x^n .

```
In[1]:= D[ x^n , x ]
```

```
Out[1]= n x-1 + n
```

Here is a slightly more complicated example.

```
In[2]:= D[x^2 Log[x + a], x]
```

```
Out[2]= 2 x Log[a + x] +  $\frac{x^2}{a + x}$ 
```

<code>D[f, x]</code>	the (partial) derivative $\frac{\partial f}{\partial x}$
<code>Integrate[f, x]</code>	the indefinite integral $\int f dx$
<code>Integrate[f, {x, xmin, xmax}]</code>	the definite integral $\int_{xmin}^{xmax} f dx$
<code>Sum[f, {i, imin, imax}]</code>	the sum $\sum_{i=imin}^{imax} f$
<code>Product[f, {i, imin, imax}]</code>	the product $\prod_{i=imin}^{imax} f$
<code>Solve[lhs==rhs, x]</code>	solution to an equation for x
<code>Solve[{lhs₁==rhs₁, lhs₂==rhs₂}, {x₁, x₂}]</code>	solution to a set of simultaneous equations for the x_i
<code>Series[f, {x, x₀, order}]</code>	a power series expansion of f about the point $x = x_0$
<code>Limit[f, x->x₀]</code>	the limit $\lim_{x \rightarrow x_0} f$

Some mathematical operations.